

High Accuracy and Presentation Attack Resistant Hand Authentication via Acoustic Sensing for Commodity Mobile Devices

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Abstract—Biometric authentication schemes, i.e., fingerprint and face authentication, raise serious privacy concerns. To alleviate such concerns, hand authentication has been proposed recently. Existing hand authentication schemes, however, use dedicated hardware, such as infrared or depth cameras, which are not available on commodity mobile devices. In this paper, we present ECHOHAND, a high accuracy and presentation attack resistant authentication scheme that complements camera-based 2-dimensional hand geometry recognition of one hand with an active acoustic sensing of the other hand. To this end, ECHOHAND plays an inaudible acoustic signal using the speaker to actively sense the holding hand and collects the echoes using the microphone. ECHOHAND does not rely on any specialized hardware but uses the built-in speaker, microphone and camera. ECHOHAND does not place more burdens on users than existing hand authentication methods. We conduct comprehensive experiments to evaluate the reliability, security, and usability of ECHOHAND. The results show that ECHOHAND has a low equal error rate of 2.45% with as few as 10 training data points and it defeats presentation attacks. The results of the user study also suggest that the required hand gestures are easy to perform, and ECHOHAND is very user-friendly with low latency.

Index Terms—Hand authentication, presentation attack, acoustic sensing, hand geometry.

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I. INTRODUCTION

BIOMETRIC authentication methods have become ubiquitous on mobile devices. However, the two widely deployed schemes, namely fingerprint and face authentication, utilize users' sensitive biological traits and have raised privacy concerns [1], [2]. To alleviate the concerns, hand authentication has been proposed as a promising method [1], [3] for the following reasons: i) hands have rich intrinsic biometric features, e.g., palmprint, hand geometry, etc., that can provide reliable authentication [4]; ii) users are less concerned with hand privacy [1], while the fingerprint is widely used for law enforcement in many countries. The face is the most sensitive information for a person, and facial recognition is even prohibited in some regions. Since 2019, Amazon has launched a contactless hand authentication payment system based on palm vein and palmprint [5], [6]. Other hand authentication solutions, such as LG Hand ID [7] and PalmID [8], have also been proposed and deployed.

Simple hand authentication can be implemented with a camera. For instance, palmprint authentications use high-resolution cameras to catch the skin patterns of a palm, such as lines, points, and texture [9], [10]. However, this approach is less accurate and vulnerable to presentation attacks, e.g., the attacker can easily spoof the system using images [11]. Other systems use specialized hardware, which is not available on commodity mobile devices, to capture sophisticated traits, such as 3-dimensional hand geometry or palm vein information. For example, GesID [12] uses a depth camera to collect hand thickness. PalmID [8] and Hand ID [7] use time-of-flight cameras [13] and infrared sensors to map out the veins under the hand skin.

In this paper, we present ECHOHAND, a high accuracy and presentation attack resistant hand authentication scheme for commodity mobile devices. ECHOHAND complements camera-based hand geometry recognition of one hand with active acoustic sensing of the other holding hand. To this end, ECHOHAND plays an inaudible acoustic signal using the speaker to actively sense the holding hand and collects the echoes using the microphone. ECHOHAND is based on the observation that the way a user holds the phone affects the phone's vibration, which results in delay and attenuation of the signal propagating through structure-borne and air-borne paths [14]. ECHOHAND does not rely on any specialized hardware but uses the built-in speaker, microphone and camera. Moreover, ECHOHAND does not place more burdens

on users than existing hand authentication methods [7], [8], [12], [15] since it senses the holding hand.

Even though acoustic sensing has been used to detect and recognize motions [16], [17], [18], [19] and faces [20], existing approaches all require the sensed objects to be at least centimeters away from the sensor for better accuracy. For instance, VSkin [16] uses acoustic sensing to identify finger motions and touch gestures at the back of the device. Chaperone [17] detects whether a user is leaving a phone to prevent phone losses. VoiceGesture [18] and Lippass [19] rely on the Doppler shifts resulting from articulator motions for liveness detection to complement voiceprint authentication. EchoPrint [20] characterizes a user's face and combines acoustic features and facial landmarks to authenticate users.

However, when authenticating a holding hand, the distance between the hand and sensor is short, making it difficult to extract the echoes reflected by the holding hand and distinguish its features. To overcome these challenges, ECHOHAND uses the Zadoff-Chu (ZC) sequence [21] as the base signal and modulates it to an inaudible frequency band. ECHOHAND distinguishes the echoes reflected by the holding hand since the signals from different paths arrive at the microphone with different delays due to different propagation speeds and path distances. To extract salient acoustic features from the separated signals, we transfer a pre-trained neural network as the generalized learning-based feature extractor.

Attack models: We consider adversaries who can conduct three types of attacks to bypass ECHOHAND: i) gesture spoofing attack, where adversaries know the victim's registered hand gesture and try to spoof the system by performing the same gesture; ii) presentation attack, in which adversaries have a picture of the victim's registered hand gesture and attempt to spoof the system using the picture; iii) mimicry attack, in which adversaries have a picture of the victim's hand gesture and also mimic the holding style of the victim. The contributions of this paper are summarized as follows:

- We present ECHOHAND, which characterizes the holding hand using acoustic sensing to complement hand geometry features from the other hand. To extract acoustic features, we design a learning-based feature extractor. We also implement a hand geometry feature extraction approach, extends state-of-art camera-based hand authentication.
- We conducted comprehensive experiments to evaluate the effectiveness of ECHOHAND under different settings and real environments, e.g., different feature sets, base models, number of training data points, low light, audible noise, different devices, periods, and hardware settings. The experiment results show that ECHOHAND can achieve a low equal error rate (EER) of 2.45% with as few as 10 training data points.
- We evaluated ECHOHAND's ability to defeat the aforementioned three types of attacks. The experiment results show that attack success rates are below 1.35%. The overhead evaluation shows that ECHOHAND is efficient with a low authentication latency of 0.59 seconds and memory usage of 83 MB.
- We conducted a user study with 20 questions to understand the participants' perceptions of biometrics on mobile

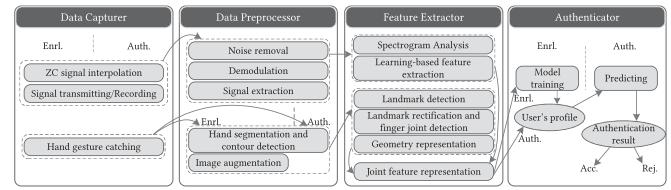


Fig. 1. The workflow of ECHOHAND.

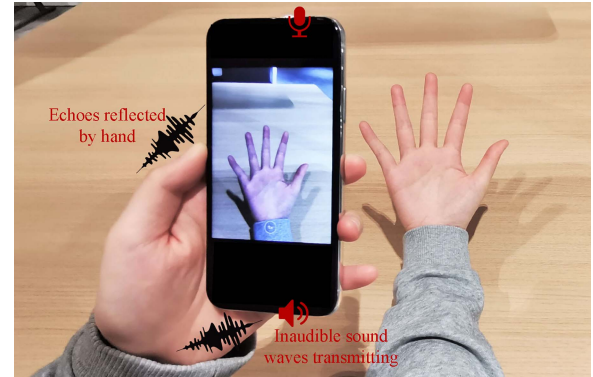


Fig. 2. Illustration of ECHOHAND.

devices and the difficulty to perform the required hand gestures. We also surveyed the recruited subjects on the usability of ECHOHAND. The results show that users' acceptance of ECHOHAND is high because it is easy to use with low latency and has few false rejections.

This paper is an extended version of our previous work [22] published in ACM Conference on Computer and Communications Security (CCS) 2022. The main differences between these two versions are listed below: i) we compare the difference of chirp and Zadoff-Chu (ZC) sequence for hand sensing and visualized comparison to help understand the difference (Section IV-A); ii) we illustrate the formulation of CWT (Section IV-C1), audio hardware selection (Section IV-A), and data collection (Section VII); iii) we conduct new experiments to evaluate the performance of transferring different base models as the feature extractor (Section VIII-B1); iv) we conduct new experiments to evaluate the performance under different numbers of training data point (Section VIII-B4); v) to measure the participants' acceptance towards biometrics on mobile device and whether it is difficult to perform the required hand gestures, we report the survey results (Section VIII-E1); and vi) to measure the usability of EchoHand, we also implemented a prototype, and recruited 30 subjects to use the prototype and then fill out another survey including four questions regarding the usage experience, e.g., if it is easy to use, authentication delay, false rejections (Section VIII-E2).

II. OVERVIEW OF ECHOHAND

As shown in Fig. 2, ECHOHAND uses the speaker and microphone for acoustic sensing to complement geometry-based hand authentication. Similar to other authentication schemes, ECHOHAND consists of two phases: enrollment and authentication. In enrollment, ECHOHAND builds a legitimate user's profile, which

includes acoustic sensed and the camera captured data. The user is required to hold the device in hand and perform a hand gesture facing the camera for registration. In the authentication phase, ECHOHAND compares both acoustic sensed and the camera captured biometrics.

As shown in Fig. 1, ECHOHAND consists of four modules: data capturer, data preprocessor, feature extractor, and authenticator. The data capturer applies interpolation to the ZC sequence and modulates it to an inaudible frequency band. ECHOHAND transmits the inaudible acoustic signal using speaker and captures the echoes using microphone. It captures the hand gesture image using the camera. For the received echoes, the data preprocessor removes noise with a band-pass filter and applies signal demodulation. ECHOHAND performs signal extraction using cross-correlation to derive the target signal shaped by the holding hand. For the hand gesture image, ECHOHAND performs segmentation to remove background and contour detection to derive the hand contour.

The feature extractor applies continuous wavelet transform (CWT) for the extracted signal to generate its time-frequency spectrogram and uses a learning-based approach to extract acoustic features. With the hand gesture image and its contour, ECHOHAND performs landmark detection and rectification to find accurate key points, and offers the hand geometry representation based on these key points. It marks the acoustic and hand geometry features as a joint feature representation. To profile the legitimate user, the authenticator uses the combined feature set to train a machine learning model based on a one-class classifier, which is later used for authentication.

III. ACOUSTIC SIGNAL PROPAGATION

When using the speaker and microphone on the same device for acoustic sensing, the received acoustic signals are a collection of the transmitted signal propagating through different paths, which is subject to the multi-path effect [23]. These paths have different time delays, phase delays, and energy attenuation thanks to the characteristics of different propagation media and distances. Multi-path propagation can be modeled as a linear time-invariant system [23], where the received signal ($y[n]$) can be described as a convolution of the input ($x[n]$) and the impulse response (IR, $h[n]$) of the multi-path propagation as shown in (1).

$$y[n] = x[n] * h[n] = \sum_{i=0}^{M-1} a_i e^{-j\theta_i} x[n - \tau_i] \quad (1)$$

where $n \in \mathbb{N}$, $y[n]$ is the n th sample in the sequence, $*$ is the convolution operator, M is the number of propagation paths, a_i is attenuation coefficient of the i th path, θ_i is phase delay, and τ_i is time delay. The IR can be formulated as (2).

$$h[n] = \sum_{i=0}^{M-1} a_i e^{-j\theta_i} \delta[n - \tau_i] \quad (2)$$

where $\delta[n]$ is the Dirac's delta function ($\delta[n] = 1$ for $n = 0$, otherwise $\delta[n] = 0$). The IR of multi-path propagation depends on many factors, such as device layout, material, the holding

hand, etc. Given a linear time-invariant system, the system output $y[n]$ can be formulated as (3).

$$y[n] = x[n] * h[n] = \sum_{k=0}^{N-1} x[k] h[n - k] \quad (3)$$

where

N is the maximum length between $x[n]$ and $y[n]$.

In particular, the cross-correlation of the output $y[n]$ and input $x[n]$ can be formulated as (4).

$$\begin{aligned} r_{yx}[\tau] &= \sum_{k=\tau}^{N-1} y[k] x^*[k - \tau] \\ &= \sum_{k=\tau}^{N-1} \left(\sum_{j=0}^{N-1} x[j] h[k - j] \right) x^*[k - \tau] \\ &= \sum_{j=0}^{N-1} \sum_{k=\tau}^{N-1} x[j] x^*[k - \tau] h[k - j] \end{aligned} \quad (4)$$

In particular, if $y[n]$ is same as $x[n]$, $r_{yx}[n]$ is also namely auto-correlation of $x[n]$, i.e., $r_x[\tau]$. Its formulation is given as (5).

$$r_x[\tau] = \sum_{k=\tau}^{N-1} x[k] x^*[k - \tau] = \sigma_x^2 \delta[\tau] \quad (5)$$

where σ_x is the standard deviation of the input signal $x[n]$. As a result, the cross-correlation $r_{yx}[\tau]$ can be expressed as (6).

$$r_{yx}[\tau] = \sum_{j=0}^{N-1} \sigma_x^2 \delta[\tau] h[\tau - j] = \sigma_x^2 h[\tau] \quad (6)$$

Thus, given that the transmitting signal $x[n]$ is constant, the $r_{yx}[\tau]$ is linearly dependent on $h[\tau]$ (6). To characterize the holding hand, we estimate the IR using the received signal and transmitted signal. The estimation of IR $\hat{h}[n]$ can be formulated as (7), which is based on the cross-correlation of the received signal $y[n]$ and the transmitted signal $x[n]$.

$$\hat{h}[n] = \frac{1}{\sigma_x^2} r_{yx}[n] = \frac{1}{\sigma_x^2} \sum_{k=n}^{N-1} y[k] x^*[k - n] \quad (7)$$

where $x^*[n]$ is the complex conjugation of $x[n]$. σ_x is the standard deviation of $x[n]$.

We use cross-correlation to measure the displacement of a signal relative to another [24]. Fig. 3(a) shows an example cross-correlation of the received multi-path signal and the transmitted signal. Because the signal propagates through the device and air, the received signal consists of the structure-borne propagation through the device and air-borne propagation. The structure-borne propagation (Path 1) arrives first due to higher speed ($>3,000$ m/s). The air-borne propagations, which consist of the direct transmission and the reflection of the holding hand (Path 2) and the reflections from other surrounding objects (Path 3), arrive later.

Fig. 4 compares the IR estimation results with two subjects in our study. Since the IR estimation is a complex-valued signal, we

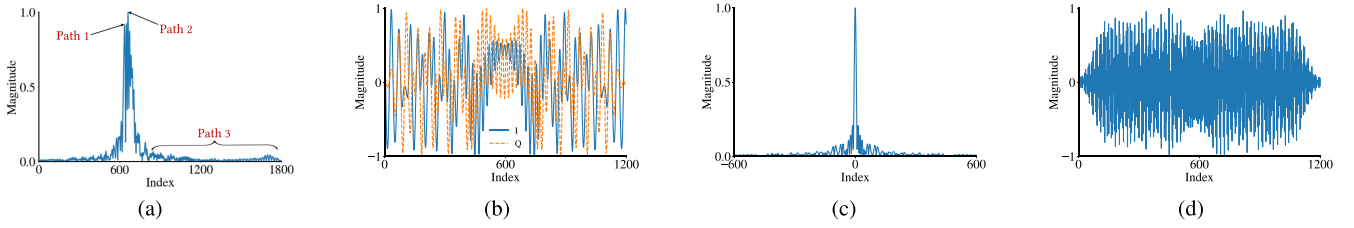


Fig. 3. An example of cross-correlation of received multi-path signal and the transmitted signal (a), real (I) and imaginary (Q) part of interpolated ZC sequence (b), auto-correlation of interpolated ZC sequence (c), and modulated sequence (d).

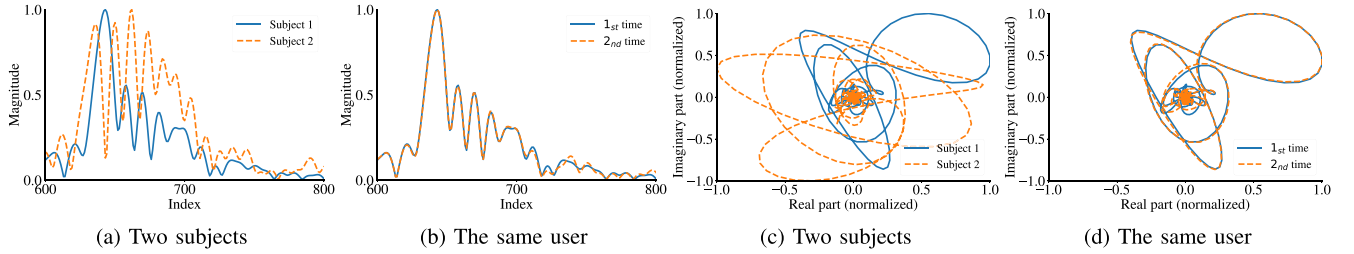


Fig. 4. IR estimations using cross-correlation of the received signal and the transmitted signal for two subjects: the magnitude of IR from two subjects (a); the magnitude of IR from the same subject at two times with 48 kHz sampling rate (b); trace of the real/imaginary parts from two subjects (c); trace of the real/imaginary parts from the same subject at two times (d).

calculate its magnitude with real and imaginary parts as shown in Fig. 4(a) and (b), and show the trace of real and imaginary parts in Fig. 4(c) and (d). As the figures show, the IR estimations of two subjects are much different. But, the two different executions from the same user are similar. Also, it shows that the trace of real and imaginary parts of two subjects are significantly different, whereas the two traces of the same user are consistent. This observation clearly shows that acoustic sensing can be exploited for characterizing hand to distinguish different subjects.

IV. CHARACTERIZING ACOUSTIC ECHOES

In this section, we present how ECHOHAND extracts acoustic features that can effectively characterize the holding hand.

A. Acoustic Data Capturer

We have the following design considerations for the transmitted signal: i) to better locate and separate the target signal reflected by hand from the received signal, the transmitted signal should have a narrow main lobe in its auto-correlation [25]; ii) the transmitted signal should be inaudible to avoid annoying users; iii) the frequency range of the transmitted signal should be supported by the frequency response of audio hardware on commodity mobile devices, i.e., less than 24 kHz.

Audio hardware selection: Common smartphones have two speakers, including a main speaker at the bottom and an earpiece speaker for phone calls at the top. Also, they equip at least two microphones to enhance voice quality. The main microphone placed at the bottom captures more voice energy. The secondary microphone is placed as far as possible, e.g., on the top, which captures less voice and more surrounding sounds. It is used for

noise cancellation by subtracting each other to get a clean voice. Considering that the signal transmitted via speaker has higher energy than the earpiece speaker, we use the speaker to play the signal. We choose the secondary microphone at the top as the receiver since the distance from the bottom speaker to the bottom microphone is extremely short, making it difficult to extract the signal shaped by the holding hand. The signal played by the bottom speaker and received by the top microphone propagates through nearly the entire device, allowing it to be shaped by the holding hand [26]. Thus, the combination of speaker and top microphone makes the holding hand characterizing more accurate.

Base signal selection: Existing acoustic sensing-based authentications [19], [20], [26] only consider magnitude characteristics by using a chirp signal, whose frequency changes with time. Chirp signal is usually used for ranging [27], [28]. We choose ZC sequence as the base signal [16], [21], which has a narrower main lobe of its auto-correlation. As shown in Fig. 3(c), the auto-correlation of ZC sequence has an extremely narrow main lobe. The auto-correlation of ZC sequence is close to zero when delay τ is not zero, whereas its auto-correlation is maximized only with a delay $\tau = 0$. To separate the multi-path signal arriving at different delays, we can perform cross-correlation of the received multi-path signal and the transmitted signal. Also, ZC sequence is a complex-valued signal with a constant amplitude, which contains not only the time-varied magnitude information but also phase information, which can help distinguish signals. A ZC sequence $x[n]$ is formulated as $x[n] = \exp(-j\pi Rn(n+1)/N)$, where N is the length of ZC sequence and $n \in [0, N-1]$, R is a constant with $R < N$, N and R are odd-valued positive integer and coprime, i.e. $\gcd(R, N) = 1$.

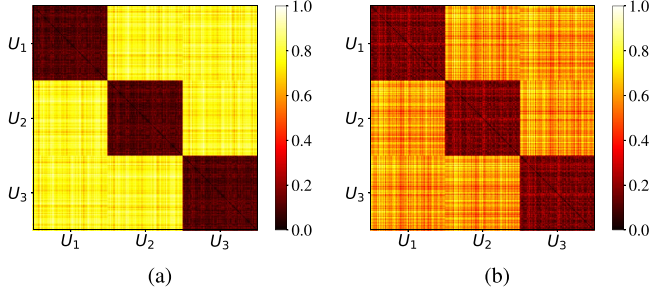


Fig. 5. The L2 distance of magnitude and phase when using chirp (a) and ZC signal (b) to sense three subjects' holding hands.

We also present the visual comparison of using chirp and ZC signal as the transmitted signal to sense the holding hand, conducting a toy example to compare the distinguishability of using chirp and ZC signal respectively for sensing the holding hand. The chirp signal sweeps from 18 kHz to 22 kHz and is with the same length of ZC sequence. We used chirp and ZC signal respectively to sense three subjects' holding hands and collected 100 acoustic echoes. For the chirp signal, we measured the L2 distance of magnitude, while magnitude and phase for ZC signal. Fig. 5 presents the L2 distance when using chirp (a) and ZC signal (b) to three subjects' holding hands. We observe that the acoustic echoes of the same subjects present consistently high similarity and the echoes of the different subjects are with consistently low similarity. While for the chirp signal, the echoes of the different subjects may show high similarity. This observation demonstrates that the ZC signal outperforms the chirp signal in characterizing the holding hand.

Interpolation: Since the raw ZC sequence covers the entire frequency band, we need to fit its bandwidth to a narrow frequency band for signal transmitting. We employ *Fourier interpolation* to increase the sequence length, which pads zeros in the frequency domain. After interpolation, we obtain the interpolated ZC sequence $x'[n]$ as shown in Fig. 3(b).

Modulation: The interpolated ZC sequence lies in an audible low-frequency band. We modulate the interpolated sequence to an inaudible high-frequency band. Considering that most smartphones only support the sampling rate up to 48 kHz, we set the sampling rate f_s as 48 kHz, and the center frequency of carrier f_c as 20 kHz. We modulate the real and imaginary probabilities of the complex-valued sequence to a single real sequence. The modulated sequence $y[n]$ is formulated as (8).

$$y[n] = \cos\left(\frac{2\pi f_c n}{f_s}\right) x'_I[n] - \sin\left(\frac{2\pi f_c n}{f_s}\right) x'_Q[n] \quad (8)$$

where $x'_I[n]$ and $x'_Q[n]$ is the real and imaginary part of $x'[n]$, respectively; $n \in [0, N' - 1]$; and N' is the length of interpolated sequence $x'[n]$. Fig. 3(d) shows an example modulated signal. A Hamming window is applied to the first and last points to reduce the audible noise caused by spectral leakage.

Fig. 6(a) illustrates signal interpolation and modulation. ECHOHAND first plays a modulated signal using the speaker and at the same time starts recording with the microphone. The playing takes N'/f_s seconds, while the recording takes

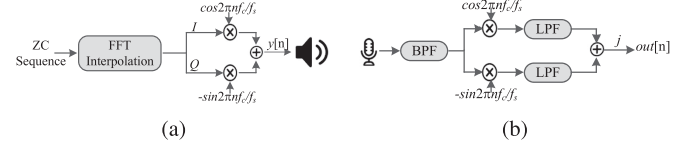


Fig. 6. Signal interpolation and modulation (a), signal preprocessing (b).

TABLE I
PROPAGATION SPEED, DISTANCE, DELAY, AND ENERGY LEVEL OF DIFFERENT PROPAGATION PATHS

Path	Speed (m/s)	Distance (cm) †	Delay (ms) / Points	Energy
1	>3,000	15.2	0.05/-19	Medium
2	~343	[15.2, 15.2×2]	[0.44/0, 0.89/22]	High
3	~343	[15.2×2, ∞]	[0.87/22, ∞]	Low

†: As an example, we use the distance of Pixel 3A in which the microphone and bottom speaker are 15.2cm apart.

$2N'/f_s$ seconds. The recorded signal is passed to the acoustic data processor module.

B. Acoustic Data Processor

The acoustic data processor performs signal preprocessing and extraction to derive the signal shaped by the holding hand.

1) Signal Preprocessor: It performs noise removal and demodulation for the captured signal to reconstruct the complex-valued baseband signal. ECHOHAND first performs noise removal to remove out-band interference. To separate high-frequency echoes from low-frequency ambient noise, we use a Butterworth band-pass filter (BPF) to filter the target signal in the transmission band. To reconstruct the baseband complex-valued signal, ECHOHAND demodulates the filtered high-frequency signal. It derives real and imaginary components by multiplying the signal and two orthogonal subcarriers that are used for modulation. We also use a Butterworth low-pass filter (LPF) to eliminate the high-frequency interference incurred by multiplication. Fig. 6(b) illustrates the flow of signal preprocessing. The output signal $out[n]$ is then used for extracting the signal shaped by the holding hand.

2) Signal Extractor: Table I presents the propagation speed, distance, delay, and energy of Path 1, 2, and 3. We aim to extract the Path 1 and 2 propagation from the received signal since they are shaped by the holding hand. We first estimate IR using cross-correlation of the demodulated signal and the transmitted signal. Then we leverage the magnitude of IR estimation to locate the candidate propagation path with the highest energy, and identify other propagation paths based on the known relative delays. Finally, we segment the target signal based on the different arrival delays of different propagation paths.

As shown in Table I, the air-borne propagation via direct transmission from the speaker to the microphone has the most energy compared with Path 1 and 3, because acoustic signal energy attenuates fast through solid device body and is absorbed more by environmental objects through the air [14]. Fig. 3(a) illustrates the energy and arrival time of three paths, the highest

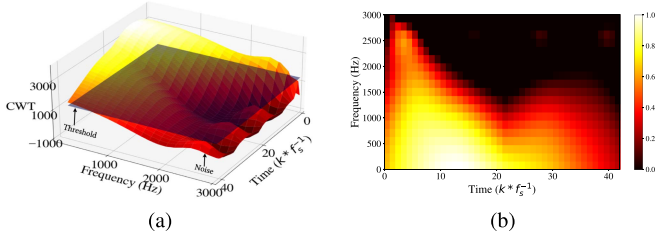


Fig. 7. An example CWT result of the magnitude: the raw CWT result (a); the CWT result after applying threshold (b).

peak represents the air-borne propagation through direct transmission. We determine the arrival time of Path 2 by detecting the highest peak.

Given the distance between the microphone and speaker as d , Path 1 arrives with $(\frac{df_s}{v_a} - \frac{df_s}{v_s})$ -point ahead of direct transmission through the air, where v_a and v_s are the propagation speed of air-borne and structure-borne, respectively. The target signal of Path 1 is within the range of $[-(\frac{df_s}{v_a} - \frac{df_s}{v_s}), 0]$, where 0 denotes the detected arrival time of air-borne propagation through direct transmission. To separate air-borne propagation signal shaped by the holding hand, we focus on Path 2 with a propagation distance ranged from d to $2d$, where d is the propagation distance of direct transmission. The target signal of Path 2 ranges from $[0, \frac{df_s}{v_a}]$ -point after Path 2. Therefore, the structure-borne and air-borne signal shaped by the holding hand is within a range of $[-(\frac{df_s}{v_a} - \frac{df_s}{v_s}), \frac{df_s}{v_a}]$ relative to Path 2. For example, when $v_a = 343$ m/s, $v_s = 3000$ m/s, $f_s = 48$ kHz, and $d = 15.2$ cm (Pixel 3A), the signal shaped by the holding hand is in the range of $[-19, 22]$ as shown in Table I. Finally, the output of acoustic data processor is the separated structure-borne and air-borne complex-valued signal, which is with the length of $(\frac{2df_s}{v_a} - \frac{df_s}{v_s})$ -point.

C. Acoustic Feature Extractor

We adopt time-frequency spectrogram analysis and learning-based feature extraction to characterize the holding hand.

1) *Spectrogram Analyzer*: For the complex-valued signal shaped by the holding hand, we calculate the magnitude and phase information. Then we employ continuous wavelet transform (CWT) [29] to construct a time-frequency representation of the magnitude and phase. We remove noise with low energy and perform normalization. CWT has better time and frequency resolution to perform time-frequency analysis than other approaches, such as short-time Fourier transform. Conceptually, the CWT is a sliding cross-correlation between a signal $f(t)$ and a family of wavelet functions, which is formulated as (9).

$$CWT(a, b; f(t), \psi(t)) = \int_{-\infty}^{\infty} f(t) \frac{1}{\sqrt{a}} \psi^* \left(\frac{t-b}{a} \right) dt \quad (9)$$

where a and b are the scales and time shifts of a reference wavelet $\psi(t)$. $*$ is the complex conjugate.

As an example, Fig. 7(a) illustrates the CWT result of the magnitude. We observe that the major components lie in a low-frequency band of 0-500 Hz, which is shaped by the user's

TABLE II
THE SIZE, NUMBER OF PARAMETERS, AND MEAN/STANDARD DEVIATION OF INFERENCE TIME TRANSFERRING DIFFERENT NEURAL NETWORKS AS A FEATURE EXTRACTOR

Base model	Size (MB)	# Of parameters	Inference time (ms)
VGG16	512.23	134,259,392	15.92/0.01
ResNet50	90.43	23,581,440	38.47/0.48
InceptionV3	83.97	21,802,208	59.53/1.37
DenseNet121	27.90	7,031,232	68.61/3.27

device holding style and the physiological characteristics of the hand. The time-frequency spectrogram also shows a significant decrease after the delay of over 30 points, which are noise components. We set the threshold as the standard deviation of the input signal to remove noise components. Fig. 7(b) shows the CWT result after applying the threshold and normalization, where the noise components have little influence on the time-frequency spectrogram.

2) *Learning-Based Feature Extractor*: To analyze the distinguishable features from the time-frequency spectrogram, we build a learning-based extractor based on transfer learning [30]. We train a base model using acoustic sensing data from different subjects, then transfer the pre-trained base model to a generic feature extractor.

Base model structure: We use a lightweight convolution neural network, DenseNet [31], as the base model. DenseNet has four dense blocks, in which the feature maps of all preceding layers and the current layer are concatenated and then passed on as the input to the subsequent layer. DenseNet recognizes input with 3 channels, whereas the time-frequency spectrogram has 2 channels, i.e., magnitude and phase spectrograms. To make it compatible with the output of spectrogram analyzer, we add a 3x3 convolutional layer before the input layer of DenseNet. Also, we add the fully-connected layer and the SoftMax layer after the output of the model to distinguish different subjects. Thanks to the lightweight base model with few parameters, its size is 27.90 MB and the average inference time is 68.61 ms, which is available on mobile devices. We have also considered other CNN structures, including VGG, ResNet, and Inception. Table II presents the comparison of using other models.

Although it has the largest inference time, this delay is short enough to support user authentication.

Base model training: We train the base model using collected data points from 15 subjects, where each subject contributes 500 data points. We use Adam optimizer for parameters optimization and categorical cross-entropy as the loss function. The training batch size and epochs are set as 100 and 5,000. We used DenseNet121 in Keras [32] as the base model and TensorFlow as the backend. Training the base model takes around two hours using 1x Tesla P40 GPU. The base model only requires to be trained only once, and then can be transferred to unseen subjects for feature extraction [30].

Transferring the base model as a feature extractor: The basic idea of transfer learning is to transfer the knowledge from a pre-trained teacher model (base model) to a new student model (feature extractor). Since the shallow layers, i.e. forward layers, have already learned representative features for the student task

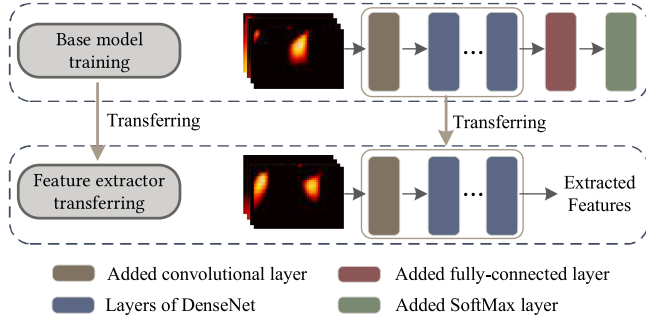


Fig. 8. Transferring the base model as the feature extractor.

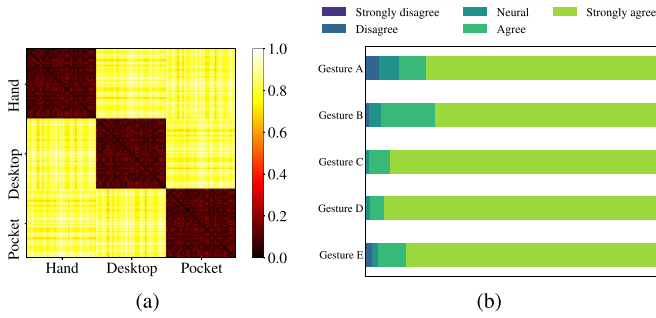


Fig. 9. L2 distance of acoustic features under three scenarios (a), participants' perception of whether the hand gesture is easy to perform (b).

during base model training, the output of these shallow layers can be used as the extracted features [20], [33]. Fig. 8 presents the illustration of transferring the base model as a feature extractor. We build the feature extractor by dropping the fully-connected layer and SoftMax layer, and saving the former layers as the feature extractor. The output is a 1024-dimensional acoustic feature vector.

We also investigated the distinguishability of using acoustic sensing to sense three typical scenarios, where the device is in the user's hand, pocket, and on a table. Fig. 9(a) shows the L2 distance of acoustic features under three scenarios, where 50 data points of each scenario are used to extract acoustic features. We observe that the features of the same scenario present a higher correlation than that of different scenarios.

V. HAND GEOMETRY FEATURE EXTRACTION

In this section, we present our hand geometry feature extraction method to help validate the effectiveness of ECHOHAND. Compared with the existing methods [34], [35], [36], our implementation contributes new methods in hand segmentation, landmark rectification, hand joint detection, and geometry representation.

A. Hand Gesture Image Processing

Hand gesture: To capture 2-dimensional hand geometry features from an image, our system imposes the following rules: i) the fingers and palm should be approximately in the same plane; ii) the fingers should be straight and not overlap with each other. These rules can be seen in most mature hand authentication

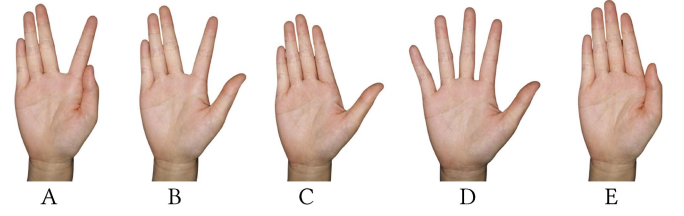


Fig. 10. Five hand gestures, where fingers and palm should be roughly in the same plane, and fingers should be straight.

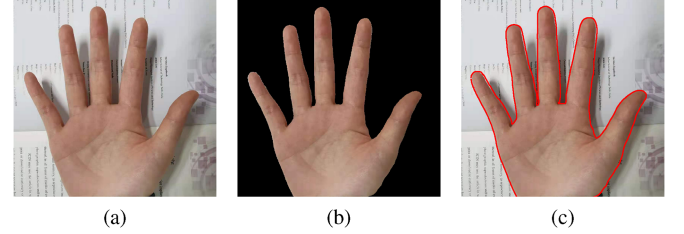


Fig. 11. An example of the original image (a), hand segmentation (b), and detected hand contour (c).

TABLE III
PARAMETERS OF IMAGE AUGMENTATION

Operation	Parameters
Scaling	Scaling factor $\in [0.8, 1.2]$
Rotation	Rotation angles $\in [-60^\circ, 60^\circ]$
Translation	Translation percent $\in [-0.1, 0.1]$
Shearing	Shearing factor $\in [-30, 30]$

systems, such as Amazon One [6] and Hand ID [7]. Fig. 10 presents five example hand gestures in our experiments. These hand gestures are easy to perform, and these restrictions will not undermine the user experience. Fig. 9(b) presents participants' perception of the difficulty of performing five hand gestures (Appendix A, available online), where 'agree' means the hand gesture is easy to perform. In enrollment, the user is required to choose a hand gesture to register, and then perform the registered hand gesture in the authentication phase.

Hand segmentation and contour detection: To eliminate the impact of cluttered image backgrounds, we utilize the DeepLabv3+ model [37] to identify the location of the hand in the image. We transform RGB to HSV color space and apply a color range to obtain the clean hand image [38]. After deriving the clean hand image, we conduct hand contour detection [39]. The detected contour is the collection of pixel location of hand edges. Fig. 11 presents an example of hand segmentation and contour detection.

Image augmentation: To generate more training data, we use image augmentation for each captured image in enrollment [40]. The idea is to generate new images that are similar to the registered hand gesture image. We consider four image augmentation operations: scaling, rotation, translation, and shearing. Specifically, we perform image augmentation using the combination of these operations. Table III presents the parameter range for these operations. As an example, Fig. 12 shows the captured

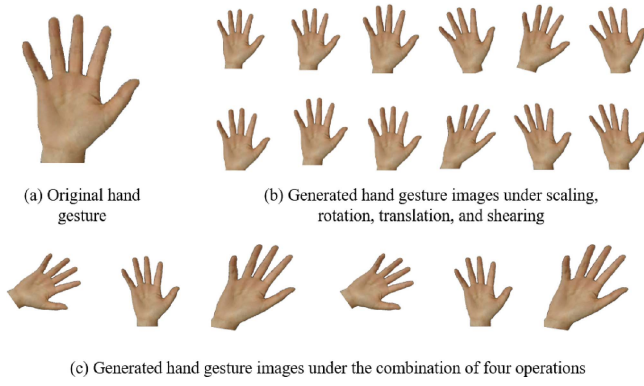


Fig. 12. Generated hand gesture images under image augmentation.

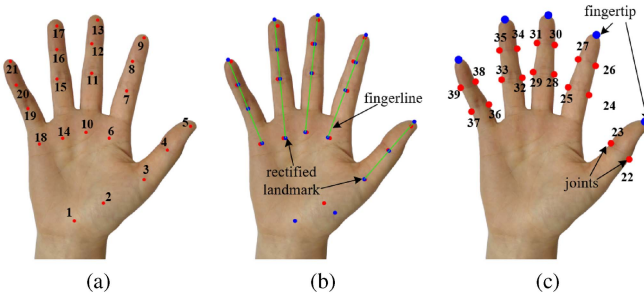


Fig. 13. The labeled red points #1-21 are the detected hand landmarks using Openpose (a), the labeled blue points are rectified landmarks based on the finger line (b), the labeled blue points are rectified fingertip landmarks and red points are detected finger joints (c).

raw hand gesture image, and generated images under different image augmentation operations.

B. Hand Geometry Representation

To extract geometry features from hand gesture images, we first detect the hand landmarks. Next, we rectify the biased landmarks and detect the finger joints. Finally, we extract hand geometry features using the rectified landmarks.

Landmark detection: To analyze the hand landmarks, we employ a released hand landmark detection model in Openpose [41], [42], which was trained to detect 21 hand landmarks.

As shown in Fig. 13(a), the labeled red points (#1-21) are detected hand landmarks, i.e., hand skeleton nodes.

Landmark rectification: Although the model has achieved great performance in detecting the hand landmarks, the estimated landmarks may not be accurate. To rectify the biased points, we consider that each finger is straight, i.e., the key points on the same finger are on the same straight line. Thus, the detected point of #6, 7, 8, and 9 should be on the same straight line, which we term as *finger line*. As shown in Fig. 13(b), we fit a straight line using the landmarks on the finger line, e.g., #6, 7, 8, and 9, based on the least squares method. Then, for each point on the finger, we find its perpendicular foot on the finger line as the rectified point, as shown as blue points in Fig. 13(b). We also rectify the fingertip point as the nearest intersection point of the finger line and hand contour. To locate the finger joints, we draw a line perpendicular to the finger line based on the rectified key

TABLE IV
LIST OF EXTRACTED HAND GEOMETRY FEATURES

Feature	Description	# Of features
Finger length	Length of each finger, including 3-5, 6-9, 10-13, 14-7, and 18-21	5
Finger width	Distance between pairs of finger joints, including 22-23, 24-25, 26-27, 28-29, 30-31, 32-33, 34-35, 36-37, 38-39	9
Palm size	Area and length of polygons consisting with lines 1-3-6-10-14-18. Distance of 1-3, 1-6, 1-10, 1-14, 1-18	7
Finger distance	Distance between 2 adjacent fingers, including 2-6, 3-7, 4-8, 5-9, 6-10, 7-11, 8-12, 9-13, 10-14, 11-15, 12-16, etc.	16

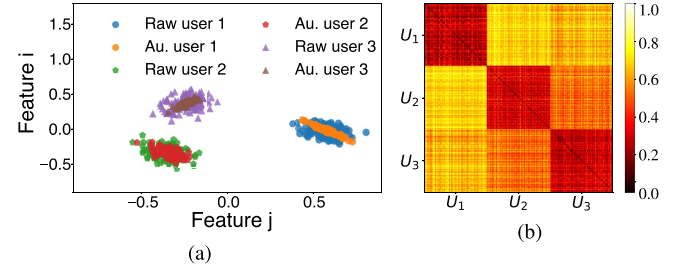


Fig. 14. Visualized feature space of raw and augmented (au.) hand gesture data under PCA (a), L2 distance of three users' hand geometry features (b).

point (blue points in Fig. 13(b)). The two nearest intersection points of the perpendicular line and hand contour are detected as a pair of finger joints (red points in Fig. 13(c)).

Geometry representation: With rectified hand landmarks and detected finger joints, we extract the following hand geometry features: i) *finger length*, which is defined as the distance between the fingertip point and the metacarpophalangeal joint, e.g., #3, 6, 10, 14, or 18. There are 5 features for finger length; ii) *finger width*, which is defined as the distance of a pair of finger joints with the hand landmark in the middle. We calculate 9 widths for 5 fingers; iii) *palm size*, which is defined as the area and length of polygons consisting of 5 landmarks, e.g., #1, 3, 6, 10, 14, 18. To describe the shape, we also calculate the distance between point #1 and each of #3, 6, 10, 14, 18. By doing so, 7 features of the palm size are calculated; iv) *finger distance* implies how users perform the hand gesture, which is relevant to the user's hand gesture behavior. We define the finger distance as the length between 2 fingers, such as #2-6 and 5-9. We calculate 16 features of finger distance.

In total, we extract 37 features, which are summarized in Table IV. Fig. 14(a) shows the two-dimensional feature space of three users' raw and augmented hand images, which can be easily classified. Fig. 14(b) presents the differences in hand geometry features extracted from three users.

VI. ONE-CLASS CLASSIFIERS

Because only the legitimate user's data is available in enrollment, ECHOHAND uses one-class classifiers, including centroid classifier (CC), local outlier factor (LOF), and one-class support vector machine (OCSVM): i) CC [43] is a distance-based classifier, which computes the distance between the test data point and the centroid of training points; ii) LOF [44] is a density-based method to recognize the outlier data point by computing the local density deviation between the test data point and its neighbors. The data point will be considered as the outlier if it has a substantially lower density than its neighbors, where the local

density deviation is estimated based on the L2 distance of its k neighbors; iii) OCSVM [45] is a distance-based classifier. It maps input data into a feature space using a kernel function and optimizes a minimal-volume hypersphere enclosing the training data. The model is defined by a centroid and radius, and any point outside the hypersphere is considered an outlier.

VII. DATA COLLECTION

We developed an Android application with the sampling rate of audio transmitting and recording as 48 kHz for data collection. We used the bottom speaker to play the signal and the top microphone to record echoes. Fig. 2 shows the position of the used audio hardware. After receiving the IRB approval from our institute, we started our data collection in March 2020. We recruited 45 subjects (Age from 18 to 38; 19 females and 26 males). We also recruited another six subjects to role-play the attacker to carry out gesture spoofing, presentation, and mimicry attacks. Before data collection, we explained to each participant the purpose of this research and the data we collect. Each subject was asked to use the smartphone for about one minute to get familiar with the device and find a comfortable and relaxing device-holding style. We compiled the following datasets.

1) *Dataset-1*: We collected only acoustic signals from 15 subjects to train the base model on a Pixel 3 A. The app transmits the inaudible and saves the recorded signal continuously. When collecting acoustic data, the subject was required to place the device down and pick it up again to hold it in a familiar and comfortable holding style, as a holding trial. For each trial, the subject was required to perform slight hand movements to capture different data. We collected 50 acoustic points for one subject in a trial, i.e., sensing the hand continuously 50 times. We collected 1,200 data points for each subject, and it took about three minutes for a subject to complete. As a result, we collected $15 \times 1,200 = 18,000$ acoustic signals for *dataset-1*.

2) *Dataset-2*: We collected hand gesture images and acoustic sensing data from 30 subjects to compile *dataset-2*. Besides holding the device in a comfortable style, a subject also used the smartphone camera to catch the hand gesture image. A subject needed to perform each hand gesture in Fig. 10 50 times. For acoustic sensing, the app collected 500 data points for each subject. During image and acoustic data collection, subjects were asked to place down the device, pick it up again, and hold the device in a familiar holding style to help get more different data samples. A subject spent ~ 12 minutes to complete this task. As a result, we collected $30 \times 500 = 15,000$ acoustic signals and $30 \times 250 = 7,500$ hand gesture images.

3) *Dataset-3*: To evaluate the effectiveness of ECHOHAND in noisy environments, we compiled *dataset-3* with the same 30 subjects. We created a noisy environment by playing the song ‘Sugar-Maroon 5’ at ~ 62 -65dB using the speaker of another smartphone, and the app collected 500 data points for each subject. As a result, we collected $30 \times 500 = 15,000$ acoustic signals in total for *dataset-3*.

4) *Dataset-4*: To evaluate the generalization of ECHOHAND on different devices, we compiled the *dataset-4* on two more

smartphones: Xiaomi6 (5.15 inches) and Redmi Note7 (6.3 inches). We divided 30 subjects into two groups that are assigned two devices for collecting acoustic data and hand images. Similar to the procedure of *dataset-2*, we collected 500 acoustic signals and 250 hand images for each subject. As a result, we collected $30 \times 500 = 15,000$ acoustic signals and $30 \times 250 = 7,500$ hand gesture images in total for *dataset-4*. Besides, we also used Samsung GALAXY On5 (a small device with 5 inches) to collect five subjects’ acoustic echoes. For each user we collected 500 acoustic signals.

5) *Dataset-5*: To evaluate the effectiveness under real settings, we compiled *dataset-5* in four different environments: *Env-1* is a quiet meeting room with a big table and chairs. *Env-2* is a noisy but empty meeting room with music playing at ~ 42 dB. *Env-3* is a noisy and crowded room with people walking and talking. *Env-4* is a quiet room but with the inaudible high-frequency sound (same as the acoustic signal for sensing) continuously playing nearby. We called back 20 subjects and divided them into four groups to collect acoustic signals in four environments respectively. Each subject is required to collect 500 acoustic signals. As a result, we collected $15 \times 500 = 7,500$ acoustic signals in total for *dataset-5*.

6) *Dataset-6*: To evaluate the authentication consistency, we compiled *dataset-6* with five subjects in four periods with an interval of one week. In the data collection of each period, we collected 500 acoustic signals and 250 hand images for each subject. As a result, we collected $5 \times 500 \times 4 = 10,000$ acoustic signals and $5 \times 250 \times 4 = 5,000$ hand gesture images in total for *dataset-6*.

7) *Dataset-7*: To evaluate the effectiveness under the low light, we compiled *dataset-7* with five subjects. To build the low light environment, we turned off the light sources and closed the curtains in a meeting room in the evening. Each subject was required to perform each hand gesture 50 times and catch the hand gesture using the smartphone camera and the flash. As a result, we collected $5 \times 50 \times 5 = 1,250$ hand gesture images in total for *dataset-7*.

8) *Dataset-8*: To evaluate the impact of using the adjacent microphone and speaker, and covering the speaker, we collected the previous five subjects’ acoustic data in the two hardware settings on Pixel 3 A. The five subjects were first required to perform hand sensing using adjacent bottom microphone and speaker (the distance is ~ 2 cm) for 500 times, then using the bottom speaker and top microphone while covering the bottom speaker for 500 times. We collected $2 \times 500 \times 5 = 5,000$ acoustic echoes.

9) *Dataset-9*: We want to evaluate if ECHOHAND can defeat the aforementioned attacks. To increase the possibility of successful attacks, we collected the acoustic sensing data and hand gesture images of six attackers. We calculate the L2 distance between the feature vector of the attackers and the previous 30 subjects. Then, we assigned each attacker five subjects as his/her targets based on the similarity. We collected the following attack datasets: i) *dataset-9a: gesture spoofing attack*. Similar to the procedure of *dataset-2*, each attacker was asked to hold the device to collect 500 acoustic signals and 50 hand gesture images per gesture. We collected $6 \times 250 = 1,500$ hand gesture

images and $6 \times 500 = 3,000$ acoustic signals for *dataset-9a*; ii) *dataset-9b: presentation attack*. The attacker was asked to present the previously recorded hand images to the data collection device using the tablet screen, and each image was only used to present only once. As a result, we collected $30 \times 250 = 7,500$ hand images for *dataset-9b*; iii) *dataset-9c: mimicry attack*. We asked each attacker to carefully observe how the attack target holds the device at a close distance ($\sim 1.5\text{m}$) and mimic the device-holding behavior. After the attacker was confident about what they observed, she/he would mimic the subject's holding behavior to collect acoustic sensing data. The attacker is allowed to perform mimicry attacks for unlimited times, but at least 50 times, where for each attack the app actively senses the holding hand 10 times. We select 50 data points with a higher probability to be accepted by the authentication model. We collected $30 \times 500 = 15,000$ data points for the *dataset-9c*.

Note that, the 15 subjects in dataset-1 did not participate in dataset-2, 3, 4, 5, 6, 7, 8. The 6 subjects as the attackers in dataset-9 do not appear in the previous 8 datasets.

VIII. EVALUATION

In this section, we report the evaluation results of the proposed system.

A. Setup

In ECHOHAND, R of ZC sequence is 63, and the length is 127. We apply 1200-point interpolation to the 127-point ZC sequence. Thus, the sequence is 25 ms, i.e. $\frac{1200}{48000}$, and the frequency band of interpolated sequence is 0–2.54 kHz, i.e. $\frac{24k}{1200} \times 127$. The modulated sequence $s[k]$ has an inaudible frequency band of 17.46–22.54 kHz. The first and last 150 points of the modulated sequence are applied to a Hamming window. For the data processor, the cutoff frequency of LPF is 3 kHz, and the passband of BPF is 17–23 kHz. We evaluate the performance of ECHOHAND using the following metrics: False acceptance rate (FAR), False rejection rate (FRR), equal error rate (EER), and area under the ROC curve (AUC). We use frequency count of scores (FCS) [46] to visualize score distributions, and FAR (i.e., attack success rate) to evaluate resistance against attacks.

B. Reliability Evaluation

To find out how distinguishable of acoustic features, we split each user's data points into training and test sets randomly, and trained the authentication model for each subject. We used the rest data points and other users' data points to evaluate the model. We performed 5-fold cross-validation to evaluate performance, and conducted grid search to find the best parameter combination for each classifier. The best parameter of LOF was $n_estimators=3$. For OC-SVM, we used radial basis function as the kernel function, and optimal parameters γ and ν were 0.19 and 0.03.

1) *Performance of Acoustic Sensing: A. Finding the optimal middle layer to transfer the base model as a feature extractor*: The base model has four dense blocks (DB) and a fully-connected (FC) layer. We used the output of each block and FC

TABLE V
THE AVERAGE EERS OF GESTURE A, B, C, D, E (FIG. 10)

Classifier		A	B	C	D	E
W. IA	CC	7.38%	6.90%	7.52%	6.48%	7.70%
	LOF	11.15%	10.88%	11.80%	10.13%	12.15%
	OCSVM	9.31%	8.83%	8.96%	8.85%	9.37%
W/o. IA	CC	6.36%	6.16%	6.38%	6.06%	6.39%
	LOF	6.89%	5.70%	6.05%	5.91%	7.24%
	OCSVM	7.49%	6.97%	8.17%	7.10%	8.78%

layer as the features to investigate how the choice of these layers influences the distinguishability. The base model was trained using 18,000 acoustic data points of *dataset-1*. We used the acoustic data points in *dataset-2* to evaluate the reliability of extracted features, where the user is profiled using 10 training data points. Fig. 16(a) reports the EERs under the feature sets extracted from different layers. The results show that using the output of 4th dense block as the acoustic features achieves the lowest EER. Fig. 16(b) presents the comparison of different classifiers using the features extracted from 4th dense block. CC achieves an average EER of 5.60%, which is much lower than LOF or OCSVM. For later experiments, we use the output of the 4th dense block as the extracted acoustic features.

B. *Performance of extracted acoustic features*: To evaluate the effectiveness of only applying acoustic sensing to distinguish between legitimate and illegal users, we used all collected acoustic data points in *dataset-2*. Each user was profiled with 10 data points. Fig. 15(a) shows ROC curves under different classifiers. The results indicate that CC achieves a lower EER. CC, LOF, and OCSVM achieve an average EER of 5.60%, 8.77%, and 9.69%, respectively. Fig. 15(b), (c), and (d) present FCS of legitimate and illegal data points, showing that prediction scores under three different classifiers have similar distributions, while the overlapping region between the scores of legitimate and illegal data points under CC is smaller than the region of the other two classifiers. This implies that the CC outperforms LOF and OCSVM in terms of the error rate.

C. *Performance of different base models*: To evaluate the performance of transferring different base models as the acoustic feature extractor, we used all collected acoustic data points in *dataset-2*. Each user was profiled with 10 data points. Fig. 18(a) shows the EER of transferring different base models to extract features. When using CC as the classifier, VGG16, ResNet50, InceptionV3, and DenseNet121 achieve the EER of 11.42%, 13.75%, 9.82, and 5.60%, respectively. The results show that DenseNet121 is with a lower EER compared with other base models.

2) *Effectiveness of Hand Geometry Features*: To evaluate the effectiveness of only applying hand geometry features to distinguish between legitimate and illegal users, we used all collected hand image data points in *dataset-2*. For each image, we applied image augmentation (IA) and generated 100 hand gesture images used for model training. We evaluated performance for five hand gestures respectively. Each user was profiled with only 10 raw data points when not using IA, and with 1,010 data points (1,000 images were generated) when using IA. Table V reports EERs of five hand gestures under different classifiers. The results show that image augmentation

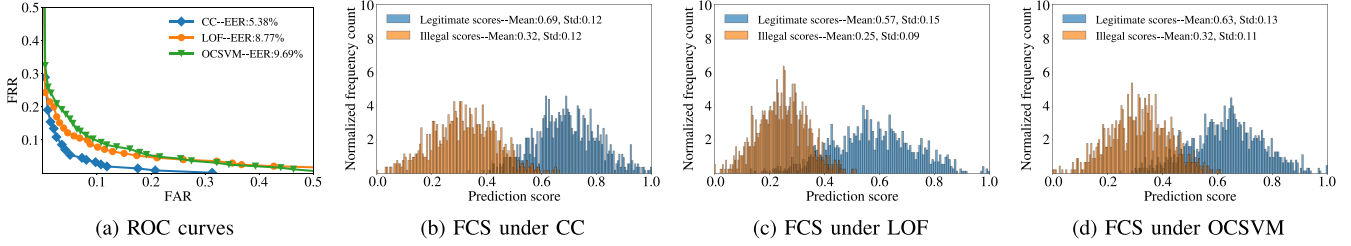


Fig. 15. ROC curves (a) and normalized FCS (b, c, d) when using only acoustic features to authenticate users.

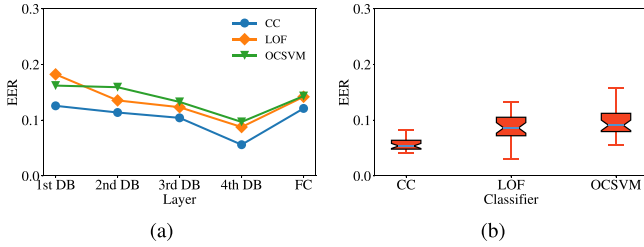


Fig. 16. Transferring different middle layers as the acoustic feature extractor: EERs under the feature sets extracted from different middle layers (a), EERs under the features extracted from the 4th dense block (b).

enhances the generalization ability of the authentication model or template, especially LOF and OCSVM. For example, the average EER of gesture A under CC, LOF, and OCSVM are 7.38%, 11.15%, and 9.31% without IA. While using image augmentation, the average EER of gesture A under CC, LOF, and OCSVM can be improved to 6.36%, 6.89%, and 7.49%.

3) *Effectiveness of ECHOHAND to Authenticate Users*: To evaluate the effectiveness of ECHOHAND, we used the collected acoustic data points and hand images in *dataset-2*. In this step, each raw image was used to generate 100 new images. Each user was profiled using 10 raw data points, i.e., 10 acoustic data points and 1,010 hand images. Fig. 17(a) shows ROC curves when using acoustic features to complement hand geometry features. CC, LOF, and OCSVM achieve an average EER of 2.45%, 5.96%, and 6.82%, respectively. Fig. 17(b), (c), (d) report FCS of legitimate and illegal data points. The results show that the overlapping region under CC is smaller than LOF and OCSVM. Besides, there exist a few illegal data points with high prediction scores under the two classifiers. The results suggest that CC outperforms LOF and OCSVM, and we used CC as the classifier due to its lower EER in later experiments.

4) *Performance Under Different Numbers of Training Data Point*: To evaluate the performance under different numbers of training data point, we used all collected acoustic data points in *dataset-2*. Each user was profiled with 1, 2, 5, 10, 15, 20, and 25 data points, where CC, LOF, and OCSVM were used as the classifier, respectively. Fig. 18(a) shows the EER under different numbers of training data point. The results indicate that more training data points contribute to a lower EER. For example, when using one data point under CC, it achieves a low EER of 6.23%, which can be improved to 1.84% if using 25 data points to profile the legitimate user. Overall, CC outperforms LOF and OCSVM if the number of training data points is less than 25.

5) *Impact of Audible Noise*: To evaluate the impact of audible noise on acoustic sensing, we profiled the legitimate user with 10 data points from *dataset-2*, and evaluated EER using acoustic data in *dataset-3* and image data in *dataset-2*, where CC was used as the classifier. Fig. 19(a) shows ROC curves under the environments with and without noise. If using acoustic features to complement hand geometry features, it achieves an average EER of 2.66% and 2.45% under the environment with and without audible noise, where the error rate is almost approximate. If only using acoustic features, it achieves an average EER of 6.12% and 5.38% under the environment with and without audible noise. Results suggest that the ambient noise has little impact on ECHOHAND for acoustic sensing.

6) *Consistency Over Time*: To evaluate the consistency of ECHOHAND over different periods, we used *dataset-6*. The 10 data points in week 1 were used to train the authentication model, while the rest data in week 1, 2, 3, and 4 were used to evaluate the EER. Fig. 19(b) shows the results under different periods of 4 weeks. The average EERs under acoustic features are 5.14%, 6.07%, 9.15%, and 15.98% in Week 1, 2, 3, and 4, respectively, while the EERs under the hand geometry and acoustic features are 2.10%, 2.13%, 3.03%, and 4.53%. We observed that hand geometry features are more consistent than acoustic features under different weeks. Combining these two kinds of features significantly improves authentication accuracy and consistency. We also found that the increasing EER of week 4 was due to a few subjects' device-holding behaviors changing dramatically.

7) *Performance on Different Devices*: To evaluate the performance of ECHOHAND on different devices, we used *dataset-1* and 4. We trained authentication models and evaluated performance using the data points from the same device. CC was used as the classifier and the legitimate user was profiled using 10 data points. Fig. 20(a) shows results under the different devices. The average EERs on Pixel 3 A, Xiaomi 6, Redmi Note7, GALAXY On5 are 2.45%, 7.24%, 3.69%, and 10.33%, respectively. The results suggest that the acoustic features on Xiaomi 6 and GALAXY On5 may undermine the distinguishability of hand features. This may be due to the short distance between the top microphone and bottom speaker.

8) *Performance Under Real Environments*: We also used *dataset-5* to evaluate the performance of ECHOHAND under real environments. We trained the authentication model using 10 data points in *Env-1*, and tested the model using the rest data in *Env-1*, 2, 3, and 4. Fig. 20(b) shows results under different environments, where we regard the results under *dataset-2* as *lab*

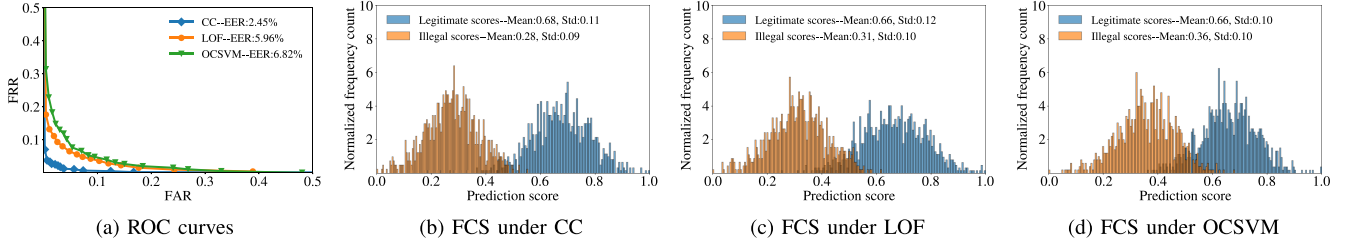


Fig. 17. ROC curves (a) and normalized FCS (b), (c), (d) when using acoustic features to complement hand geometry features.

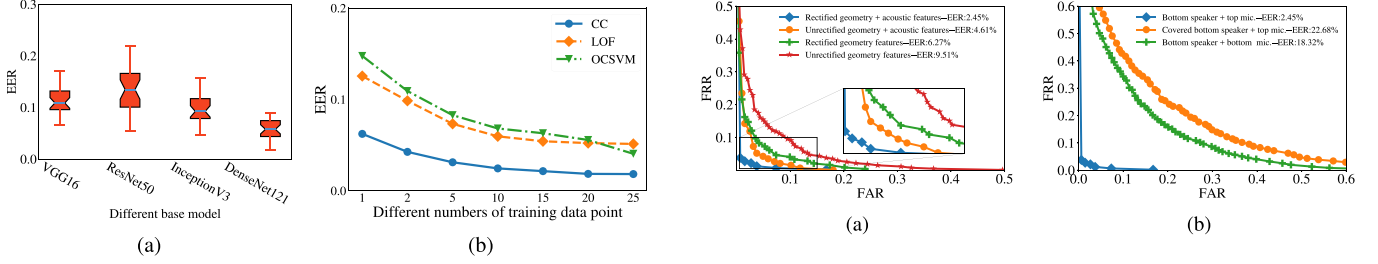


Fig. 18. EER of transferring different base models to extract features (a), EER under different number of training data points (b).

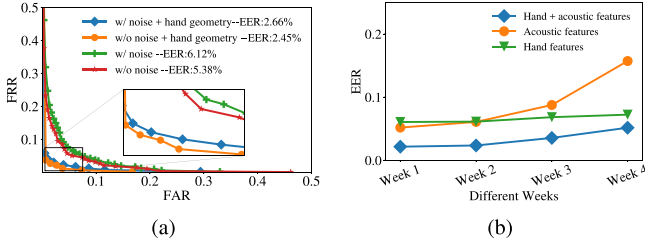


Fig. 19. ROC curves under environments with (w/) and without (w/o) audible noise (a), EERs under different weeks (b).

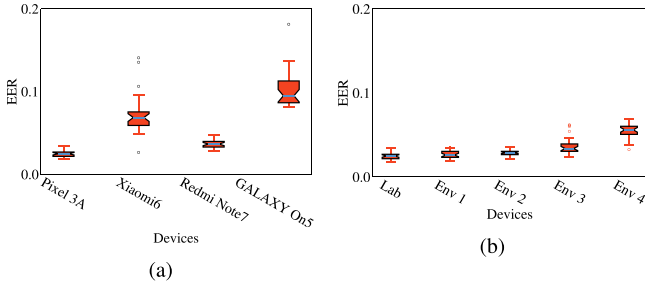


Fig. 20. EERs under different devices (a), and environments (b).

environment. The average EERs under lab and five real settings are 2.45%, 4.95%, 4.79%, 5.55%, and 6.53% respectively. The performance of five real settings is approximate to the lab setting. The results suggest that ECHOHAND is reliable in real environments.

9) *Effectiveness of Landmark Rectification*: To evaluate the effectiveness of our designed landmark rectification, we used *dataset-2*. Each user was profiled with 10 data points and CC as the classifier. Fig. 21(a) shows ROC curves under landmark rectification. If only using geometry features, the average EERs under rectified and unrectified geometry are 6.27% and 9.25%.

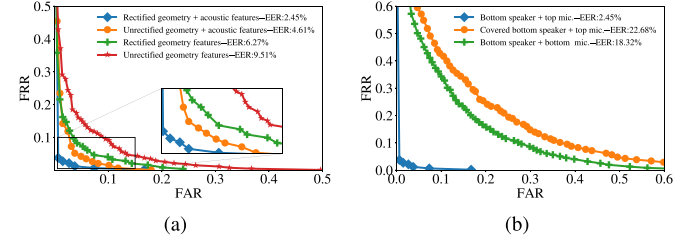


Fig. 21. ROC curves under landmark rectification (a), and different hardware settings (b).

TABLE VI
EERs AND AUC UNDER LOW LIGHT

Setting	Features	EER	AUC
Low light	Hand geometry	8.13%	0.9761
	Sensing + hand geometry	2.92%	0.9955
Normal light	Hand geometry	6.27%	0.9864
	Sensing + hand geometry	2.45%	0.9977

If using acoustic sensing features to complement hand geometry features, it achieves an average EER of 2.45% and 4.61% under rectified and unrectified geometry. The result suggests our designed hand landmark rectification improves the robustness of hand geometry features in authenticating users.

10) *Performance Under Low Light Setting*: To evaluate the robustness under low light setting, we trained the authentication model using *dataset-2*, and evaluate the authentication model using *dataset-2* and 7 respectively. Table VI shows EERs and AUC under low light. If using only hand geometry features, the EERs are 8.13% and 6.27% under low and normal light. While using acoustic features and hand geometry features, ECHOHAND achieves an average EER of 2.92% and 2.45% under low and normal light. This shows that acoustic sensing enhances the robustness of hand geometry-based authentication under low light.

11) *Impact of Different Audio Hardware Settings*: To evaluate the impact of different audio hardware settings, we used *dataset-2* and *dataset-8*. 10 data points in *dataset-2* were used to train the authentication model, while the combinations of acoustic data in *dataset-8* and hand images in *dataset-2* were used to evaluate EER. Fig. 21 presents ROC curves under different audio hardware settings. If using the covered bottom speaker and top microphone, ECHOHAND achieves an average EER of 22.68%. While using the bottom speaker and bottom microphone, the average EER is 18.32%.

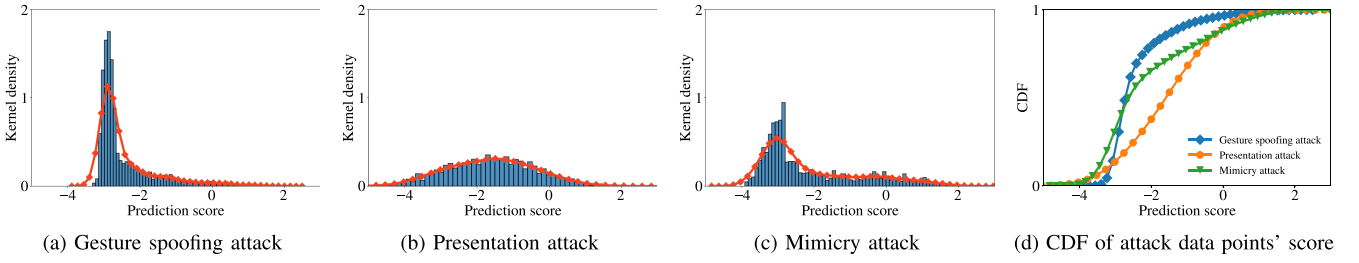


Fig. 22. PDF/FCS (a), (b), (c) and CDF (d) under gesture spoofing, presentation, and mimicry attack.

TABLE VII

FAR AND MEAN/STANDARD DEVIATION OF ATTACK DATASET'S PREDICTION SCORES

Attack type	FAR	Prediction scores
Gesture spoofing attack	0.21%	-2.42/ 0.86
Presentation attack	0.62%	-1.60/ 1.21
Mimicry attack	1.35%	-2.11/ 1.37

TABLE VIII

THE LATENCY OF DIFFERENT MOBILE AUTHENTICATION SCHEMES [30], [47]

Scheme	Latency	Motion behavior ¹
PIN	1.25s	✓
Pattern lock	3.14s	✓
Fingerprint authentication	0.29s	✗
Facial authentication	1.48s	✗
ECHOHAND	0.59s	✗

¹ Require users to perform motion behavior, e.g., typing, and drawing.

C. Evaluation of Attack Resistance

To evaluate the resistance against three different attacks, we used *dataset-9a*, *b*, and *c* to test the authentication model trained using *dataset-2* from the previous 30 subjects. We normalized the decision threshold (where FAR=FRR) to 0, and then investigated the distribution of the attack dataset's prediction scores. We reported FAR, i.e., attack success rate, distribution of the attack dataset's prediction scores, the kernel density of prediction scores evaluated under Gaussian kernel [30], and the cumulative distribution function (CDF).

Fig. 22(a) shows the prediction scores' distribution under gesture spoofing attack using *dataset-9a*. The mean prediction score of attack data points is -2.42. The kernel density shows a narrower range but consistently low prediction scores. ECHOHAND can defend against gesture spoofing attack with a FAR of 0.21%. Fig. 22(b) shows the prediction scores' distribution under presentation attack using hand images of *dataset-9b* and acoustic data of *dataset-9a*. The mean prediction score is -1.60. The kernel density shows a wide range but a higher score than the scores under the gesture spoofing attack. Presentation attack is with a FAR of 0.62%. Fig. 22(c) shows the prediction scores' distribution under mimicry attack using hand images of *dataset-9b* and acoustic data of *dataset-9c*. The mean prediction score is -2.11. The kernel density shows a wide range but is with consistently low prediction scores, where there exist many attack data points with the scores ranging from -4 to -2. Mimicry attack is with a FAR of 1.35%.

The results suggest that ECHOHAND can defeat gesture spoofing, presentation, and mimicry attacks. Table VII reports FAR and prediction scores of attack data points. Fig. 22(d) presents the CDF of prediction scores, where the scores less than zero are with a high probability.

D. Latency and Memory Usage

The authentication latency is composed of the time required for data processing, feature extraction, and model inference. We evaluated the latency of these modules respectively and

monitored the total memory usage. We developed a prototype system of ECHOHAND on Android, and evaluated the average latency for 50 authentications. It takes about an average latency of 0.37, 0.14, and 0.08 seconds for the 3 modules respectively on Pixel 3 A (2x2.0 GHz). In total, ECHOHAND requires 0.59 seconds to complete authentication. Table VIII compares the latency between ECHOHAND and different mobile authentication schemes. We also used Android Profiler to monitor the memory usage of ECHOHAND, and the average memory usage on Pixel 3 A is 83 MB.

E. User Study

We also perform two user studies to measure difficulty to perform hand gesture and the system usability respectively.

1) *Users' Experience on Hand Gestures*: Our user study was conducted based on 20 questions in our user study.¹ Finally, we collected 194 valid answers for the questionnaire, where question 10 is used to decide whether an answer is valid. Table X presents the demographics of the participants. Table IX reports participants' sentiment of biometrics on mobile devices. Generally, fingerprint is more prevalent than other biometrics. Hand authentication is the potential popular authentication among biometrics. Fig. 23(a) shows participants' perception of biometrics on mobile device. The results show that users shows major concern on the leakage of biometric data, and prefer hand authentication to other biometrics regarding privacy leakage. Fig. 23(b) shows participants' perception of the difficulty to remember and perform the hand gesture again. The results suggest that the hand gesture is easy to perform and remember.

2) *User Study on System Usability*: We also implemented a prototype and asked the previous 30 subjects to fill out another survey including four questions regarding the usage experience of ECHOHAND. The results are summarized in Table XI. 22 of 30 subjects strongly agree that ECHOHAND is easy to use,

¹<https://www.wenjuan.com/s/BvUFBvx/>

TABLE IX
PARTICIPANTS' SENTIMENT OF BIOMETRICS ON MOBILE DEVICES

	Female		Male	
	No.	Rate	No.	Rate
Q3) What authentication method do you use most frequently on your primary smartphone?				
PIN/Pattern lock	59	2%	43	8%
Fingerprint recognition	40	39%	16	13%
Face recognition	28	11%	5	6%
Others	2	12%	1	3%
Q4) Which of the following biometrics do you prefer on your primary smartphone?				
Fingerprint	58	30%	29	15%
Face	22	11%	6	3%
Hand	40	20%	26	13%
Iris	2	1%	1	1%
Voiceprint	4	2%	1	1%
Gait	1	1%	0	0%
Signature	2	1%	2	1%
Q5) For the widely deployed biometrics, e.g., fingerprint/facial recognition, which of the following aspects do you concern most?				
Security	39	26%	28	14%
Usability	39	26%	9	5%
Privacy	51	20%	28	14%

TABLE X
OVERALL DEMOGRAPHICS OF THE PARTICIPANTS

	Female		Male	
	No.	Rate %	No.	Rate %
Age range	129	66%	65	34%
18 or younger	3	2%	15	8%
18-24	76	39%	25	13%
25-30	21	11%	11	6%
31-40	24	12%	6	3%
41-50	4	2%	4	2%
51-60	1	1%	4	2%

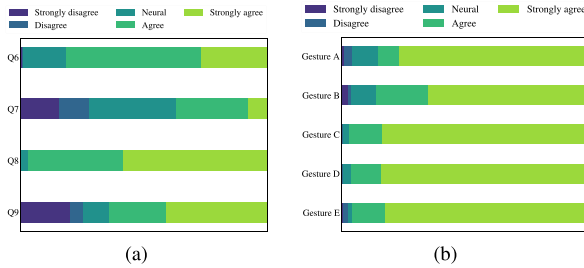


Fig. 23. Participants' perception of biometrics on mobile device (a), and difficulty to remember and perform the hand gesture again (b).

and 19 subjects strongly agree that the delay of ECHOHAND is acceptable. Generally, the survey results imply that most subjects are willing to use ECHOHAND as the authentication method, since they think it is easy to use, the system delay is acceptable, and there are only few false rejections.

IX. RELATED WORK

Hand authentication: Hand authentication schemes distinguish the legitimate user and impostors based on the intrinsic hand traits. Palmprint-based methods rely on the high-resolution camera to capture palmprint (i.e., a complex set of skin lines), and sophisticated image processing methods to extract the texture features [10], [48]. Nevertheless, it is vulnerable to presentation attacks [9], [11]. Palm and finger vein-based methods rely on the dedicated hardware, e.g., an infrared camera, to scan the pattern of blood vessels [6], [7], [13], [49], [50], [51]. The dedicated hardware is not available on most commodity mobile devices.

Hand geometry-based methods identify the legitimate user by analyzing the hand geometry features [12], [15], [34], [35], [52], such as finger lengths, and widths. Simple methods relying on a monocular RGB camera to analyze 2D hand geometry are usually vulnerable to presentation attacks [34], [35], [36]. To enhance security, these methods are dependent on comprehensive camera systems, such as depth camera [12]. Some behavior-based hand authentications also characterize the hand/finger movements to identify different subjects [12], [15], [52], [53], [54], [55]. However, these methods require the user to perform predefined hand gesture movements.

Compared with hand authentications that require users to perform hand motion hand movement [12], [15], [52], ECHOHAND does not require users to perform any hand gesture movement but make a stationary hand gesture. ECHOHAND can also resist presentation attacks. Table XII presents the comparison of existing mature commercial hand authentications and the latest research work.

Acoustic sensing-based authentications: Acoustic sensing exploits speakers and microphones to design fancy applications [16], [17], [59], [60], [61], [62], [63], [64], [65], [66], [67], e.g., usable and secure authentications [18], [19], [20], [68], [69], [70], [71]. Sound-proof [68] and Proximity-Echo [72] were proposed to validate whether the user's phone is near the device used to log in via sensing ambient noise. Echoprint [20] enhanced the security of face authentication against presentation attacks by transmitting an inaudible acoustic signal and receiving the echo to sense the facial geometry. To enhance voiceprint against replay attacks, VoiceGesture [18] and Lippass [19] leveraged active acoustic sensing to detect mouth movements to validate the presence of the legitimate user. To maintain the security of pattern locks and PINs, TouchPrint [67], [69] leveraged acoustic sensing to characterize finger tapping/sweeping events. To provide hand authentication using acoustic sensing, Echolock [26] considered only the structure-borne signal for sensing and analyzed acoustic features in the time and frequency domain. To protect smartphone notification privacy, [59] leveraged vibration response from microphone and accelerometer spectrogram to identify the user's hand gripping device, where it relies on audible message tones and cannot work well in motion scenarios. ECHOHAND is different in that it silently senses and profiles a user's hand using the sound propagating through the device

TABLE XI
PARTICIPANTS' SENTIMENT OF EXPERIENCE ON ECHOHAND

Questions	Strongly agree	Agree	Neutral	Disagree	Strongly disagree
Q1: ECHOHAND is easy to use.	22	5	2	1	0
Q2: The delay of ECHOHAND is acceptable.	19	7	3	1	0
Q3: ECHOHAND seldom reject authorized user.	22	7	1	0	0
Q4: I am willing to use ECHOHAND as the authentication method.	21	5	3	1	0

TABLE XII
COMPARISON OF EXISTING MATURE COMMERCIAL HAND AUTHENTIFICATIONS, THE LATEST RELATED RESEARCH WORK

Method	Required hardware	Description of hand features	EER	PAR ¹	Hand motion ²
Commercial product					
Amazon One [7]	Unknown customized hardware	Palm vein and palmprint patterns	N/A	✓	✗
Hand ID [8]	Infrared illuminator, TOF sensor ³	Palm vein patterns	N/A	✓	✗
PalmID [9]	Infrared camera	Palm vein patterns	N/A	✓	✗
PalmID [9]	RGB camera	Palmprint patterns	N/A	✗	✗
PalmSecure [56]	Near-infrared imaging camera	Palm vein patterns	N/A	✓	✗
Vein ID [51]	Near-infrared illuminator, common RGB camera	Finger vein patterns	N/A	✓	✗
Research paper					
[13]	Leap motion controller ⁴	3D motion depth features of gesture movement	~ 2%	✓	✓
[52]	Leap motion controller	3D motion characteristics of fingertips and finger joints	< 4%	✓	✓
[16]	Multi-touch screen	Hand geometry and motion characteristics of swiping on a multi-touch touchscreen	5.84%	✓	✓
[35]	Optical scanner	Hand geometry features, including finger width and length	0.59%	✗	✗
[34]	Optical scanner	Hand geometry graph topology	3.05%	✗	✗
[57]	RGB camera, infrared lamp	Palm dorsal veins and hand geometry features	1.87%	✓	✗
[58]	IntelRealSense ⁵	Palm vein patterns	< 1%	✓	✗
[36]	RGB camera	Hand images features extracted from different layers of a neural network	~ 5.2%	✗	✗
[26]	Speaker, microphone	Time-domain, frequency-domain, MFCC ⁶ , and chromagram features of structure-borne echos when holding a device (Without solid hand features)	~ 6%	✓	✗
[59]	Speaker, microphone, accelerometer	Spectrogram of microphone and accelerometer incurred by notification tones when holding a device (Without solid hand features)	~ 5%	✓	✗
ECHOHAND	RGB camera, speaker, microphone	Learning-based acoustic features of structure-borne and air-borne echos while sensing the hand holding device, hand geometry features including finger length, width, palm size and finger distance	~ 2.45%	✓	✗

¹Presentation attack resistant. ²Require users to perform hand motion. ³A type of depth camera with a range imaging camera system. ⁴An infrared-based depth camera used for tracking motions.

⁵A high quality LiDAR-based depth cameras. ⁶Mel-frequency cepstral coefficients, a kind of typical acoustic features.

and air to provide reliable and secure hand authentication on off-the-shelf devices.

X. DISCUSSION

Although we took great efforts to maintain our studies' validity, there are limitations in our studies and experiments.

Small device: ECHOHAND may not work on devices where the distance between the microphone and the speaker is extremely short, which makes extracting the signal shaped by the holding hand difficult. When transferring the feature extraction model of Pixel 3 A to a small device, Samsung GALAXY On5 (a small device with 5 inches), the EER increases from 2.45% to 10.33%.

Other unusual usages: ECHOHAND achieves a competitive EER of 2.45%, and while direct comparison with fingerprint or facial recognition is challenging due to different sensing modalities, it offers clear advantages in scenarios such as low light, wet fingers, and presentation attacks. However, ECHOHAND has its own limitations. It requires the user to hold the device in hand during authentication, making it unsuitable when the device is placed on a desk or stand, such as during meetings. Additionally, users wearing gloves may not be authenticated successfully. Since hand landmark detection relies on vision algorithms, it shares limitations with facial recognition, being sensitive to poor lighting and off-angle views, which can lead to higher false rejection rates in such conditions.

Behavior variability: A user's phone-holding style may change over time, leading to potential false rejections and performance degradation, as observed in Week 4 of our evaluation. This highlights the challenge of behavioral drift in behavioral biometrics like ECHOHAND. Inspired by mechanisms used in systems like FaceID, future versions of ECHOHAND could incorporate periodic re-enrollment or lightweight model

updates to adapt to such changes while maintaining usability and accuracy.

Sampling rate: A lower sampling rate may undermine the time resolution of the target signal separating, and lead to the low discernibility of extracted acoustic features among different users. Besides, the transmitting signal should be carefully designed again to meet the goal of the transmitting band. For example, when setting the supported sampling rate on commodity devices, e.g., 8 kHz, 11.025 kHz, 16 kHz, 22.05 kHz, 44.1 kHz, and 48 kHz, the extracted signal (shaped by the holding hand) has 8, 11, 14, 20, 38, and 42 points respectively. A smaller number of points may lead to worse system performance.

Limited evaluation scenarios: In our evaluation, EchoHand was tested on data collected from 30 subjects across 18 representative settings, including normal usage, audible noise, low light, four time periods, four devices, four real-world environments, and three audio hardware configurations. While this setup is more comprehensive than prior work, it still does not encompass all real-world factors. Additionally, the user population falls within a limited age range, which may not capture variations in hand geometry and phone-holding behavior due to age-related changes or usage habits. Future work will involve larger-scale and more demographically diverse studies, along with refined feature designs, to better assess EchoHand's generalization and ensure long-term scalability and robustness.

User feedback from our study indicated that EchoHand was generally perceived as convenient and fast to use. However, some participants raised concerns about microphone access, highlighting the need for transparent data practices and user trust. EchoHand activates the microphone only briefly during authentication and does not require continuous audio monitoring, which helps minimize privacy concerns. To further address

this, future versions may incorporate privacy-preserving mechanisms such as on-device processing, permission prompts, and clear communication about when and how acoustic sensing is used.

In addition, while hand gesture authentication has been explored in prior work, we incorporate it in EchoHand not as a primary modality but as a complementary cue to enhance system robustness. Our experiments show that phone-holding behavior alone provides strong discriminative features, but incorporating coarse-grained gesture information improves resistance to mimicry and spoofing attacks. This fusion helps balance accuracy and security without relying on complex or specialized gesture recognition techniques.

XI. CONCLUSION

In this paper, we present a high accuracy and presentation attack resistant hand authentication method for commodity devices. It uses built-in hardware on off-the-shelf devices, including the microphone, speaker, and camera. To mitigate the threats of presentation attacks, it characterizes the holding hand using acoustic sensing techniques to complement the hand geometry features. We compiled nine datasets to evaluate the reliability and security of ECHOHAND. The evaluation results demonstrate that ECHOHAND is robust and can identify the legitimate user and impostors with a low EER.

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